

FRIDAY OGOCHUKWU IKWUOGU

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RESEARCH PROFILE

Computer Science researcher with interdisciplinary academic and industry experience in computer networks, software-defined networking (SDN), artificial intelligence (AI), cloud-native networking, and enterprise network infrastructure. Research focuses on AI-assisted SDN and 5G/6G network architectures, network performance optimization, and system-level analysis through mathematical modeling, simulation, and experimental evaluation, with particular interest in intelligent communication systems that improve network scalability, reliability, security, and resource efficiency for next-generation wireless and distributed networks.

Integrates principles from computer networking, machine learning, distributed systems, and cybersecurity to address emerging challenges in software-defined and cloud-native communication infrastructures. Through peer-reviewed publications, academic service as an IEEE Access reviewer, institutional leadership within IEEE, and enterprise networking experience across critical energy and telecommunications infrastructure, seeks to bridge theoretical research with practical deployment challenges in modern communication systems.

Long-term research goal: to contribute to the design of intelligent, resilient, and autonomous communication networks that leverage AI-driven optimization to improve the performance and security of future wireless and software-defined networking environments.

RESEARCH INTERESTS

Primary Research Areas	Secondary Research Areas
Computer Networks	Network Security and Resilience
Wireless Mobile Networks	Zero Trust Network Architecture
Software-Defined Networking (SDN)	Network Automation and Orchestration
AI-Assisted Network Management	Federated Learning for Communication Networks
5G/6G Communication Systems	Post-Quantum Network Security
Network Performance Modeling and Analysis	
Network Optimization	Cloud Computing
Cloud-Native Networking	Internet of Things (IoT)
Edge and Distributed Computing	Critical Infrastructure Communications
Intelligent Communication Systems	

EDUCATION

Master of Science in Computer Science

University of Texas Permian Basin, USA | Expected: May 2024 | GPA: 4.00/4.00

Relevant Coursework: Networks, Artificial Intelligence, Machine Learning, Cloud Computing, Advanced Database Systems, Cybersecurity, Software Engineering, Data Structures and Algorithms

Selected Graduate Projects:

- Scalable Cisco-Based Enterprise VoIP Network with Redundant Infrastructure
- CorpView: Cloud-Based Business Intelligence and Data Visualization Platform
- Data Encryption and Decryption Performance Analysis
- Secure Enterprise Network Design
- Distributed Cloud Computing Applications

Master of Science in Information Technology

National Open University of Nigeria | May 2019 | GPA: 3.85/4.0

- Research Project: Predictive Analytics for Network Performance Monitoring Using AI Techniques

Bachelor of Science in Electrical & Electronic Engineering

University of Ibadan, Nigeria | March 2011

- Senior Design Project: Design and Implementation of an Automated Rain-Gauge System

RESEARCH EXPERIENCE

Graduate Research Assistant — University of Texas Permian Basin, USA Aug 2022 – May 2024

Conducted interdisciplinary research in computer networking, cloud-native systems, cybersecurity, and distributed computing while supporting faculty-led research and IT initiatives.

- Conducted comprehensive literature reviews on software-defined networking (SDN), cloud-native architectures, artificial intelligence, and next-generation communication systems.
- Performed system modeling, performance analysis, and technical evaluation of networked and distributed computing environments.
- Investigated AI-assisted approaches for improving network performance, infrastructure resilience, and intelligent system management.
- Assisted in designing experimental methodologies for evaluating communication systems and cloud-based infrastructure.
- Supported SQL-based data management and analysis for research projects involving enterprise information systems.
- Prepared technical reports, research documentation, and contributed to peer-reviewed scholarly manuscripts.
- Collaborated with faculty and multidisciplinary research teams on networking, cloud computing, and cybersecurity projects.
- Applied Python and SQL for data processing, automation, and research-related analysis.

Research Areas: Computer Networks, Cloud Computing, Software-Defined Networking, Artificial Intelligence, Cybersecurity, Distributed Systems, Network Performance Analysis

PUBLICATIONS

Peer-Reviewed Journal Publications

- Infrastructure-as-Code for 5G RAN, Core and Service-Based Interface Deployment: A Comprehensive Review.** *Global Journal of Engineering and Technology Advances*, 21(3), 144–167, Dec 2024. Second Author.
- Securing Software-Defined Networks (SDN) Against Emerging Cyber Threats in 5G and Future Networks: A Comprehensive Review.** *International Journal of Engineering Research & Technology (IJERT)*, Mar 2025. Third Author.
- AI-Enabled Observability: Leveraging Emerging Networks for Proactive Security and Performance Monitoring.** *International Journal of Innovative Research and Scientific Studies*, May 2025. Fifth Author.
- AI-Enhanced Predictive Maintenance Systems for Critical Infrastructure: A Cloud-Native Architecture.** *World Journal of Advanced Engineering Technology and Sciences*, Nov 2024. Fourth Author.
- Federated IAM for Critical Systems: A Blockchain-Based Decentralized Identity Framework Enhanced by Federated Learning for Cross-Sector Critical Infrastructure Trust.** Fifth Author.
- Decentralized AI-Driven Zero-Trust Architecture: Leveraging Blockchain for Immutable Policy Enforcement and Autonomous Anomaly Response in Critical Infrastructure Systems.** Fourth Author.
- NLP-Driven Zero Trust: Automating Security Policy Generation and Access Review via Semantic Analysis of Operational Manuals in Critical Infrastructure.** Third Author.
- Quantum-Resilient Infrastructure: Migrating U.S. Financial Payment Systems to Post-Quantum Cryptography Standards to Prevent “Harvest Now, Decrypt Later” Attacks.** First Author.
- Adversarial MLOps and the Protection of the Agentic Attack Surface in Distributed Autonomous AI Systems.** Fifth Author.

Research Impact

Metric	Value
Peer-Reviewed Publications	9+
Research Citations	182
Research Reads	1,862

Research Recommendations	9
Research Interest Score	115.8
IEEE Access Reviews Completed	80+

Research Themes

- Artificial Intelligence for Communication Networks
- Software-Defined Networking (SDN)
- AI-Assisted 5G/6G Network Architecture
- Network Performance Optimization
- Cloud-Native Network Infrastructure
- Zero Trust Architecture
- Federated Identity Management
- Network Security
- Critical Infrastructure Communications
- Post-Quantum Security

PEER-REVIEW AND EDITORIAL SERVICE

Active Peer Reviewer — IEEE Access

2024 – Present

Serve as a peer reviewer for IEEE Access, evaluating manuscripts spanning artificial intelligence, wireless communications, computer networking, distributed systems, cybersecurity, and intelligent communication technologies.

Areas of Review:

- Software-Defined Networking (SDN)
- 5G/6G Wireless Communications
- Network Performance Optimization
- AI for Communication Systems
- Federated Learning
- Graph Neural Networks
- Reinforcement Learning
- Computer Vision
- Internet of Things
- Cybersecurity
- Cloud Computing
- Edge Computing
- Intelligent Transportation Systems
- Medical AI
- Signal Processing

Academic Service:

- Completed reviews for 80+ research manuscripts, delivering rigorous assessments of research methodology, innovation, technical contribution, and scientific impact.
- Evaluated research involving network optimization, wireless communication systems, AI-driven networking, federated learning, and distributed computing.
- Provided technical recommendations that contributed to editorial decision-making and manuscript improvement, aligned with IEEE research standards and publication ethics.

Institutional Leadership & Promoting the Profession

- IEEE South Plains Section — Educational Activities Leader: direct regional engineering education programs, technical curriculum review, and STEM advancement initiatives.

- Professional Mentorship: developed workshops mentoring early-career engineers toward global competency frameworks such as the BCS Chartered IT Professional (CITP).

SELECTED RESEARCH PROJECTS

AI-Assisted Network Performance Optimization

Investigated AI-driven approaches for improving communication system performance, resource allocation, and network observability in next-generation wireless networks.

Infrastructure-as-Code for 5G Networks

Conducted a comprehensive review of Infrastructure-as-Code methodologies for automated deployment and orchestration of 5G Radio Access Networks (RAN), Core Networks, and Service-Based Interfaces.

Secure Software-Defined Networking

Reviewed emerging cyber threats affecting SDN-based 5G architectures and analyzed mitigation strategies for secure network operation.

Enterprise Cisco VoIP Network

Designed and implemented a scalable Cisco-based enterprise Voice over IP network featuring redundant routing, switching, and fault-tolerant architecture.

CorpView Enterprise Analytics Platform

Designed and developed a cloud-based enterprise business intelligence platform integrating database management, visualization, and reporting for organizational decision support.

Network Performance Monitoring Using Artificial Intelligence

Developed predictive analytics approaches for monitoring communication network performance using artificial intelligence techniques during graduate research.

Data Encryption Performance Analysis

Evaluated computational performance and security trade-offs among modern encryption algorithms for enterprise environments.

Survey of Data Security in Fog-Cloud Computing-Based IoT

Conducted a comprehensive literature review of security mechanisms, privacy challenges, and architectural considerations for fog-cloud computing environments.

PROFESSIONAL EXPERIENCE

IT Client Support Specialist (Contract) — EPMS – Energy Transfer

Sept 2025 – Present

Midland, Texas

Provide enterprise IT and identity-management support within a large-scale critical energy infrastructure environment, supporting network-connected systems, endpoint devices, and enterprise applications.

- Support enterprise network infrastructure involving TCP/IP, DNS, DHCP, VPN, Active Directory, and Microsoft Entra ID.
- Lead enterprise identity and access management initiatives strengthening secure access across corporate and operational technology (OT) environments.
- Direct cybersecurity governance through access controls, endpoint security strategies, and vulnerability management practices that enhance the resilience of critical infrastructure.
- Standardize technology lifecycle management and compliance processes aligned with IEC 62443, NIST, and corporate security standards.
- Collaborate with cybersecurity, infrastructure, and vendor teams to resolve complex technical problems and maintain operational continuity.

IT Field & Infrastructure Engineer (Contract) — Synergy Consultant – Marathon Petroleum

May 2024 – Aug 2025

Midland, Texas

Supported enterprise networking and communication infrastructure across multiple operational facilities, including integration of legacy operational technology (OT) with modern cloud-enabled infrastructure.

- Led the design and deployment of secure, resilient enterprise network infrastructure across geographically distributed operational sites.
- Installed and configured Cisco routers, switches, wireless infrastructure, and enterprise communication systems; participated in rack-and-stack deployments and structured cabling.
- Served as primary IT lead in cross-functional collaboration with engineering teams on infrastructure modernization projects.
- Supported Microsoft 365, VPN connectivity, and enterprise software deployments.

Technical Support Analyst / IT Technical Analyst (Contract) — **Ovation Workplace Inc. – ExxonMobil** June 2024 – April 2025

Midland, Texas

Provided Tier II and Tier III support for enterprise networking, communication systems, and digital workplace services.

- Led resolution of complex Tier III enterprise incidents, improving operational stability and reducing mean time to resolution (MTTR) through ServiceNow-driven incident management and analytics.
- Directed Microsoft 365 identity modernization initiatives and implementation of secure remote access solutions.
- Resolved incidents involving Active Directory, Microsoft Entra ID, VPN connectivity, and enterprise endpoints; collaborated with network engineering teams on routing, switching, and firewall issues.

Graduate Research Assistant (IT & Systems Support) — **University of Texas Permian Basin** Aug 2022 – May 2024

Odessa, Texas

Supported campus network systems while participating in research involving distributed systems, cloud computing, and enterprise information systems.

IT Security Officer — **Petroleum Training Institute**

2015

Provided strategic solutions for enterprise cybersecurity, IT infrastructure, and information security governance across the institution.

- Led the design, procurement, and deployment of enterprise-grade next-generation firewall solutions, strengthening the institution's cybersecurity posture.
- Directed development of enterprise security policies, compliance programs, and cybersecurity awareness initiatives that enhanced regulatory compliance and reduced organizational risk.

Network Engineering Intern — **Nokia Siemens Networks**

Mar 2009 – Dec 2010

Lagos, Nigeria

- Supported communication infrastructure implementation and maintenance, including routing, switching, network documentation, and hardware lifecycle management.
- Assisted in the design and deployment of core network infrastructure using routing protocols (BGP, OSPF, EIGRP, VRF) to support carrier-grade network availability.
- Supported multi-layered secure connectivity frameworks across regional telecommunication backbones.

TECHNICAL SKILLS

Programming: Python, SQL, PowerShell

Computer Networks: TCP/IP, Software-Defined Networking (SDN), Cisco Routing and Switching, VLANs, VPN, DNS, DHCP, Enterprise Network Architecture, Network Troubleshooting, Wireless Networking, BGP/OSPF/EIGRP/VRF

Cloud and Distributed Systems: Microsoft Azure, Microsoft Entra ID (Azure AD), Google Cloud Platform, IBM Cloud, Docker, Cloud-Native Infrastructure, Windows Server, Linux

Enterprise & Security Technologies: Microsoft 365, Active Directory, ServiceNow, Endpoint Management, Systems Administration, IEC 62443, NIST Frameworks, Zero Trust Architecture

Research Skills: Network Performance Modeling, Experimental Evaluation, Literature Review, Technical Writing, Research Methodology, Data Analysis, System Modeling, Performance Analysis

PROFESSIONAL CERTIFICATIONS

- Cisco Certified Network Associate (CCNA Routing & Switching)
- ITIL Foundation Certification

- IBM Introduction to Cloud Computing
- Google Cloud Platform Fundamentals: Core Infrastructure
- Python Programming
- Using Python for Automation
- Microsoft Azure Networking Concepts
- Azure DevOps Fundamentals
- Docker on Azure
- Learning ServiceNow
- Technical Support Fundamentals (Coursera)
- Health Informatics and Technology (UTHealth Houston)
- Advanced Networking (Delta State Ministry of Science and Technology)

PROFESSIONAL MEMBERSHIPS

- Institute of Electrical and Electronics Engineers (IEEE) — Senior Member
- IEEE Computer Society — Senior Member
- IEEE Young Professionals
- British Computer Society (BCS) — Professional Member, UK
- Institution of Engineering and Technology (IET) — Member, UK
- Nigerian Institution of Professional Engineers and Scientists (NIPES) — Fellow Membership, Nigeria
- Council for the Regulation of Engineering in Nigeria (COREN) — Licensed to Practice
- Nigerian Society of Engineers (NSE) — Corporate Member
- Society of Petroleum Engineers (SPE) — Professional Member
- National Society of Black Engineers (NSBE) — Professional Member, USA

CONFERENCES AND PROFESSIONAL DEVELOPMENT

Conferences

34th Council for the Regulation of Engineering in Nigeria (COREN) Engineering Assembly — 2026

Theme: Advancing Public Safety in Nigeria through Strategic Engineering Regulation, Enforcement, and a Tiered Sanctioning Regime

- Participated in technical sessions on engineering innovation, public safety, and infrastructure development.
- Engaged with engineering professionals on emerging technologies and regulatory best practices.

Nigerian Society of Engineers (NSE) International Engineering Conference — 2025

Theme: Engineering Innovation for a Sustainable Blue Economy

- Participated in discussions on emerging engineering technologies, digital transformation, and sustainable infrastructure.
- Explored advances in communication technologies and engineering innovation.

Professional Development

IEEE Continuing Education — “AI Unpacked: What Leaders Need to Know to Stay Ahead” — 2025

Participated in professional development focused on artificial intelligence, emerging technologies, and their impact on engineering, industry, and digital transformation.

Continuing Professional Education

Completed specialized training in: Cloud Computing, Microsoft Azure, Google Cloud Platform, Cisco Networking, Python Programming, Docker, Azure DevOps, ServiceNow, Health Informatics, and Enterprise IT Infrastructure.

ENTREPRENEURIAL EXPERIENCE

Founder & Managing Director — SchoolGrade Link (Educational Technology Startup)

2017 – Present

- Founded and led a technology startup focused on cybersecurity education, digital workforce development, and enterprise technology solutions.
- Provided strategic leadership in designing and delivering cybersecurity awareness and digital skills initiatives that strengthened workforce capabilities and cyber resilience.
 - Directed development of industry-aligned cybersecurity training programs and enterprise technology solutions, enhancing technical competency, data protection, and organizational digital transformation.
 - Led enterprise application planning and architecture, business process automation, database design, and product development.
- Co-Founder — ABBI Elite Forum (Non-Profit Organization)** 2017 – Present
- Co-founded and provided strategic leadership for a professional development and networking platform focused on community advancement, collaboration, and knowledge exchange.
- Designed and implemented community engagement frameworks and educational outreach initiatives fostering professional growth, leadership development, scholarship, and technology-driven advancement across diverse disciplines.

MEDIA CONTRIBUTIONS

- Invited technical opinion articles and press coverage discussing emerging technologies, digital infrastructure, and cybersecurity:
- BusinessDay Nigeria — “AI and ML in Public Safety: Enhancing Security in Nigerian Cities”
 - BusinessDay Nigeria — “Building a Digital-First Economy: The Pathway to Nigeria's \$1 Trillion GDP Goal on the African Continent”
 - BusinessDay Nigeria — “Quantum Risk Is a Global Risk: Why the World's Financial Systems Must Prepare Now”
 - BusinessDay Nigeria — “Digital Infrastructure and IoT Innovation in Nigeria's IT & Energy Industry”
 - ThisDay Nigeria — “Where Cybersecurity Meets Energy Security”
 - Daily Trust — “Pioneering Digital Resilience: Contributions to Enterprise Technology and Cybersecurity in Nigeria”

AWARDS, SCHOLARSHIPS, AND HONORS

Graduate Research Assistantship — University of Texas Permian Basin (UTPB)
August 2022 – May 2024

- Awarded a competitive Graduate Research Assistantship in the Department of Computer Science in recognition of academic excellence and research potential. Supported faculty research, enterprise information systems, and university technology initiatives while pursuing graduate studies.
- Maintained a 4.00/4.00 GPA throughout the M.S. Computer Science program.
 - Conducted interdisciplinary research in computer networks, cloud computing, distributed systems, and artificial intelligence.
 - Assisted faculty with research activities, technical documentation, and scholarly publications.

TEACHING AND MENTORING EXPERIENCE

- Research Mentor — Data Analytics — Tomorrow's Leaders Today (TLT), Inc.** *Spring 2024 – Spring 2025*
- Mentored undergraduate interns participating in research and data analytics projects, fostering analytical thinking, research methodology, and professional development.
- Mentored interns in data collection, cleaning, analysis, and interpretation.
 - Guided students in research design, literature review, and evidence-based problem solving.
 - Assisted interns with preparing technical reports and research documentation.
 - Promoted ethical research practices, teamwork, and professional communication.

- Teaching Interests:**
- Computer Networks
 - Data Analytics
 - Artificial Intelligence

- Cloud Computing
- Research Methodology
- Database Systems
- Enterprise Information Systems